

WILLIAM CHIU

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Education

University of Waterloo

2024 - Present

Bachelor of Software Engineering, Honours Co-Op
Colonel Hugh Heasley Scholarship Recipient

Waterloo, ON | CGPA: 3.93/4.00

Work Experience

Waterloo Vision & Image Processing Lab

May 2026 – Present

Undergraduate Research Assistant

Waterloo, ON

- Demonstrated a **skeleton-based action recognizer's** robustness to pose noise on a **27K-clip NHL dataset**, with $|r| < 0.04$ between pose confidence and accuracy on 5,383 clips via body vs. stick keypoint statistical tests.
- Improved hockey action recognition mean per-class accuracy from **20.7% → 49%** by fusing a **skeleton GCN** with a **Conv1D motion encoder**, recovering rare motion-defined classes such as backward skating from **~0% → 81%**.

Blair AI

January 2026 – April 2026

Software Engineer

Toronto, ON

- Saved **7+ staff hours per week** in manual fax-filing by building a multimodal AI fax classification platform processing **60,000+ documents annually**, with document rotation correction and text sharpening, Avaros EMR integration, and Terraform-managed AWS deployment.
- Improved **OCR prompt performance** by **35%** across **1,000+ labeled faxes** by building an **LLM-driven prompt refinement and benchmarking platform** with **concurrent evaluation**, **automated labeled-data ingestion**, **stratified sampling**, **metric summarization**, and **historical backtesting**.
- Won **6-figure annual contracts** from major hospitals by extending the fax-processing platform into an end-to-end **MRI referral system** with a scalable multi-tenant **Postgres** model, Terraform-managed **AWS SES/SNS** communication pipelines, and **voice AI** integration for autonomous follow-up calls

Wat.AI: SEE-DR

September 2025 – March 2026

Deployment and Modeling Team

Waterloo, ON

- Evaluated **Swin-UNet** and **CMAC-UNet** for diabetic retinopathy lesion segmentation through model comparison
- Improved lesion segmentation robustness to class imbalance by developing a composite **Tversky+Dice** loss

Ontario Government (MPBSDP - GSIC)

May 2025 – August 2025

Full Stack Developer

Toronto, ON

- Modernized **CI/CD** with a **Kubernetes**-based architecture and templated **Azure DevOps** pipelines, achieving **zero-downtime deployments** for the organ donor registration platform.
- Led development of the Organ Donor Registration backend, improving system reliability through a **Java Spring Boot** architecture, **C# API Manager** integrations, and a **WebClient circuit breaker** for failed API calls.
- Reduced email template retrieval latency by **80%** via **Caffeine caching**, and protected sensitive data for **16M+ Ontarians** through an **AOP logging** framework with a JSON data masker.

Projects

Polyhedral Order Book | C++20, CMake, Catch2, HiGHS

- Implemented a **polyhedral order book** in C++20, generalizing order crossing to N assets via **primal/dual LPs** solved with HiGHS, enabling **multi-leg trades** and **correlated-basket market making**.
- Validated correctness across **crossed/uncrossed books**, **partial fills**, and **multi-level trades** against a baseline **2-asset CLOB** with a **Catch2** test suite under **CMake/Ninja**.

FitWithDyce.com | TypeScript, NextJS, Vercel, Twilio, Resend, Supabase

- Shipped a **full-stack fitness platform** from 0 to 1 with **TypeScript**, **NextJS**, **Google OAuth**, **Twilio SMS**, and **Resend** email for automated client communication
- Engineered a **multi-tenant Supabase backend** with **RLS** and **REST APIs** to log workouts, chat with coaches, schedule meetings, and track nutrition

Quantum Computing Algorithm to Minimize Power Loss | Python, D-Wave

- Minimized power loss in energy systems** through a cloud-based quantum optimization program by designing the **DFS-based search strategy** and deriving the **edge-weighting formula** for the optimization algorithm.

Technical Skills

Languages: C/C++, Java, Python, JavaScript, TypeScript, HTML/CSS, SQL, Scala, VHDL

Libraries/Frameworks: React, Node.js, Express, NextJS, FastAPI, SpringBoot, NestJS, Three.js

Developer Technologies: Git, Supabase, Postman, Docker, Kubernetes, Azure, AWS, Swagger, Vercel, Railway